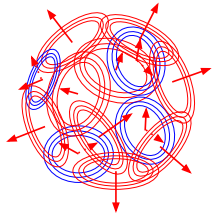
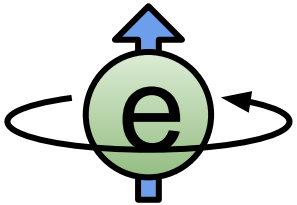
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



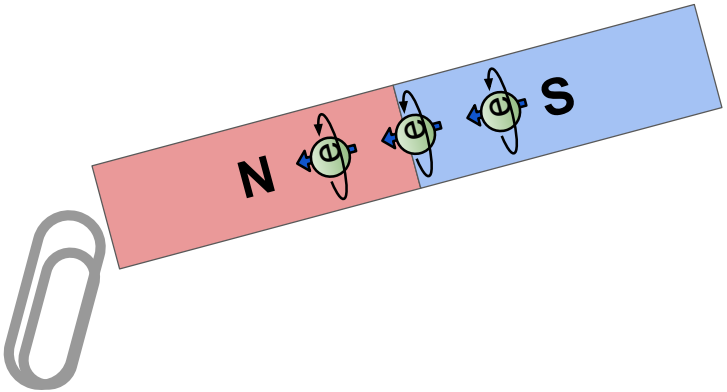
Chap.2: What is Electron Spin?

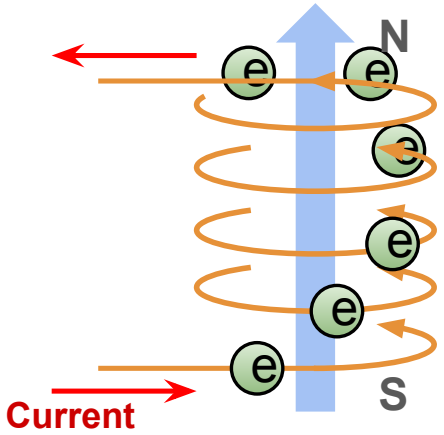


1. What is Electron Spin?

The magnetic force of a magnet is caused by the spin of electrons.

In an electromagnet, magnetic force is generated when an electric current flows through a coil; as shown in following figure, electrons possessing an electric charge rotate along the coil to produce this force.



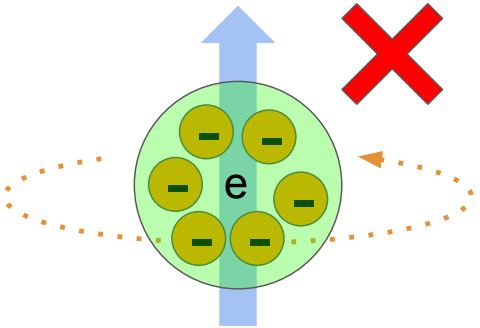


1. What is Electron Spin?

Historically, it was proposed that the internal charge of an electron rotates due to its self-rotation, similar to an electromagnet.

However, calculations showed that the rotation speed would need to exceed the speed of light, so this idea is currently rejected.

In modern quantum mechanics, quantum spin is defined as the intrinsic angular momentum of a quantum.

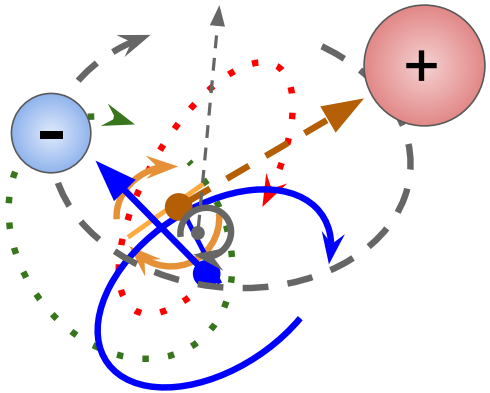


1. What is Electron Spin?

In R-space Theory, electron spin is explained by rotation vectors and their precession.

Precession is the quantum spin itself.

Furthermore, it is not the charge that rotates; rather, the combination of the rotation vector and its precession constitutes the electric charge.

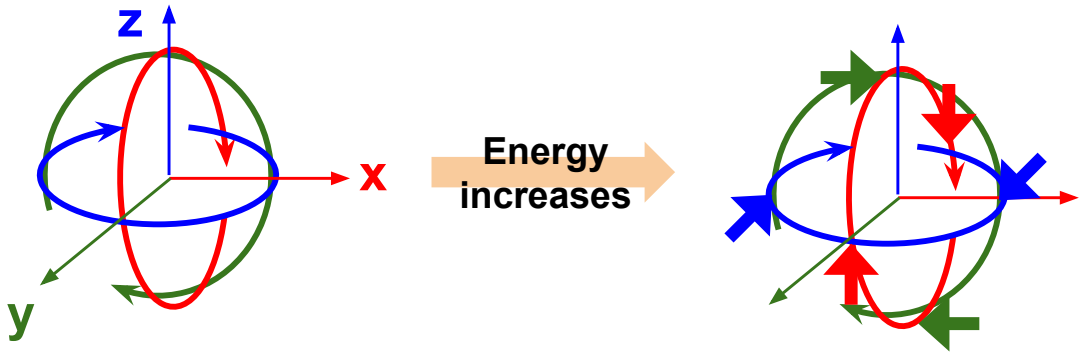


2. The Mechanism of Rotation Vector Precession

In R-space Theory, a photon consists of rotation vectors—energies possessing three orthogonal rotational force components in the x, y, and z directions.

When the photon's energy increases (e.g., during pair production), the rotation speed of these vectors increases, causing them to influence one another.

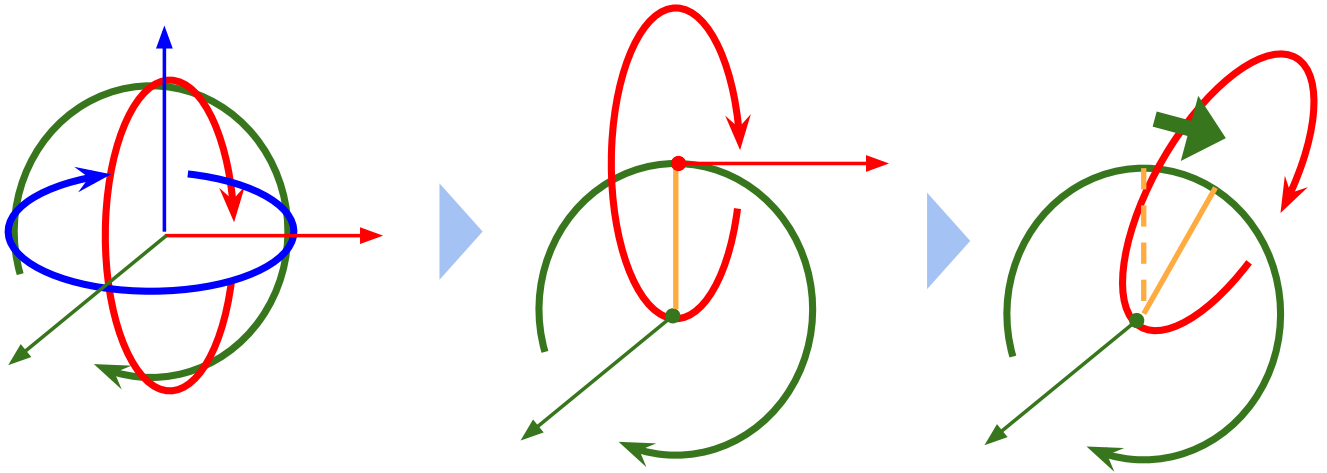
As shown in following figure, the phase of each rotation vector collides with the rotational plane of the orthogonal vectors, acting to rotate those planes.



2. The Mechanism of Rotation Vector Precession

For example, if we place the starting point of rotation vector x on the phase circle of rotation vector y, it forms a geometry like a Hopf link.

First, rotation vector x rotates slightly along the phase circle of rotation vector y.

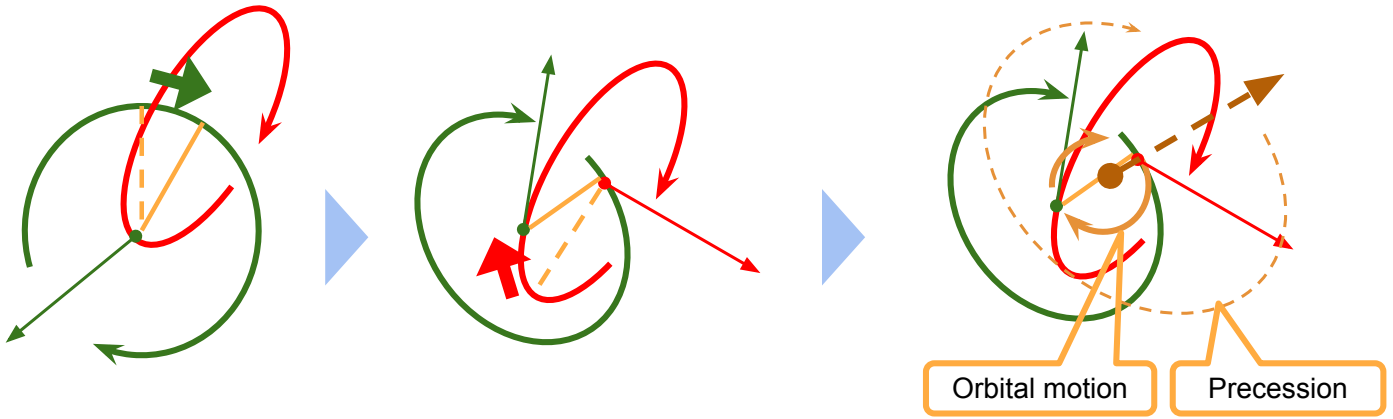


3. Orbital Motion and the 45-Degree Angle

Next, rotation vector y rotates slightly along the phase circle of rotation vector x.

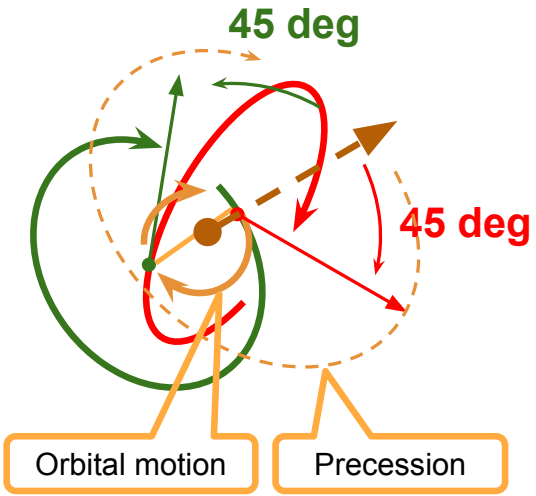
As this repeats, rotation vectors x and y orbit each other like binary stars while maintaining their orthogonal relationship.

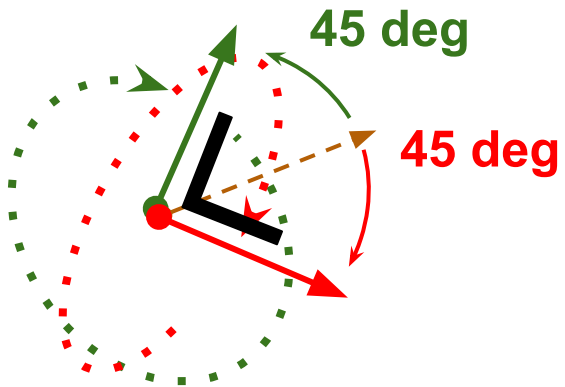
This orbital motion is the precession.



3. Orbital Motion and the 45-Degree Angle

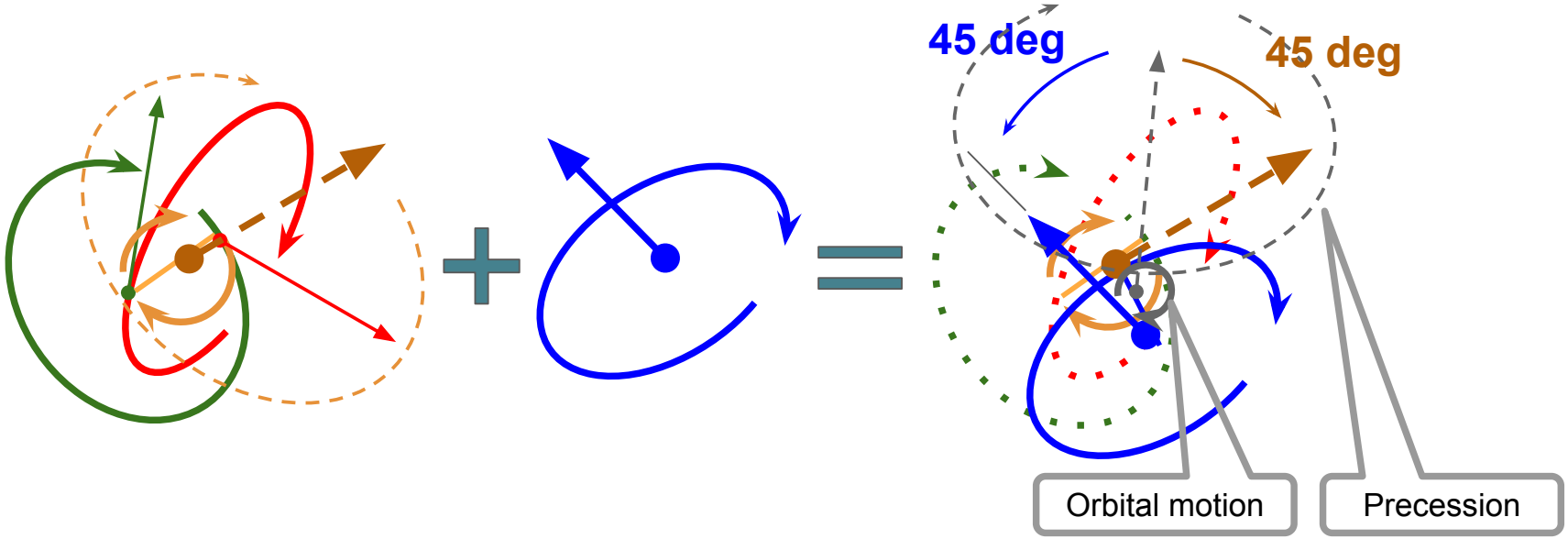
Since the vectors are orthogonal, the precession angle is 45 degrees relative to the central axis of rotation.





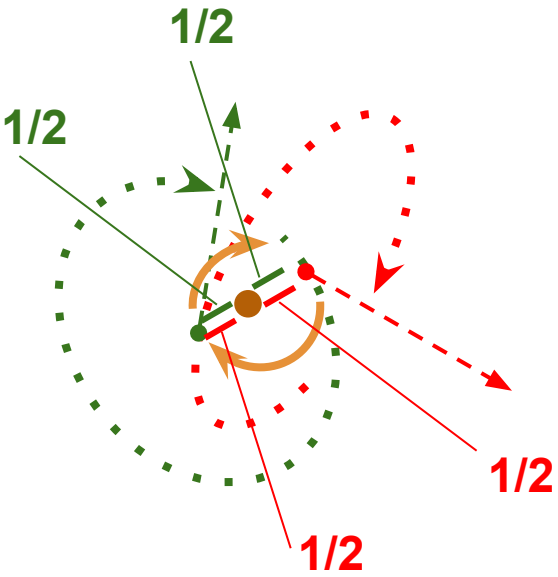
3. Orbital Motion and the 45-Degree Angle

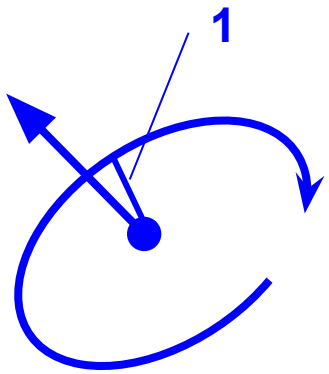
Rotation vector z then pairs with the orbital phase of the x-y pair and precesses at a 45-degree angle.



3. Orbital Motion and the 45-Degree Angle

While "size" in the conventional sense does not exist in R-space, if we treat the rotation vector as angular momentum and the radius of the phase circle as the arm length (radius 1), the arms of x and y are divided into two relative to the orbital axis, making each length 1/2.

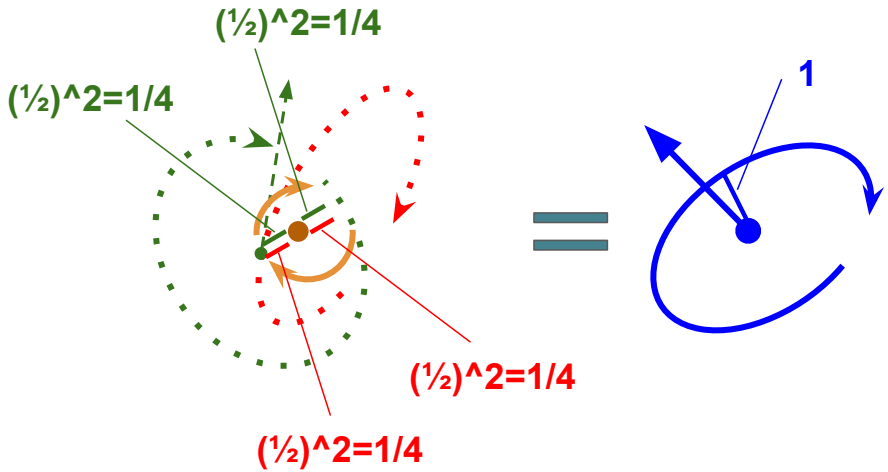




4. Balance of Angular Momentum

Since angular momentum (kg · m^2/s) is proportional to the square of the length, the four arms (two from x, two from y) of length 1/2 each have an angular momentum of (1/2)^2=1/4.

The total angular momentum of these four arms equals 1, which balances with the angular momentum of rotation vector z (which is 1).

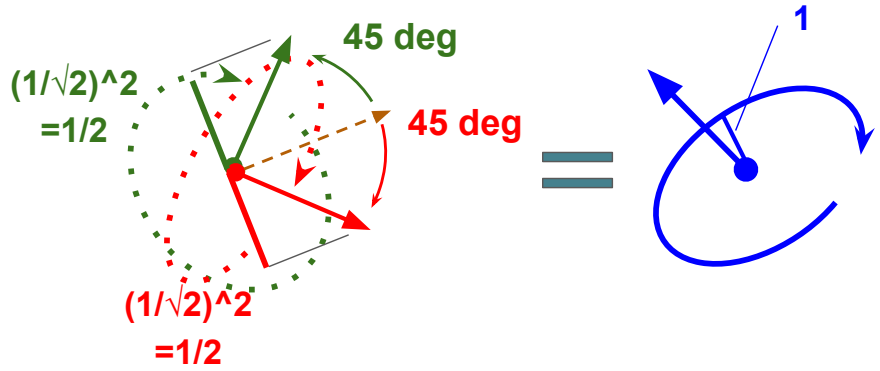


4. Balance of Angular Momentum

Alternatively, calculating as vectors, x and y are tilted at 45 degrees to the precession axis, making their arrow lengths in the orbital plane one divided by the square root of two each.

Squaring these gives 1/2 each; combined, they equal 1, balancing with vector z.

These precessing rotation vectors are electrons and quarks, and the invariance of the speed of light is due to this fixed 45-degree precession angle.

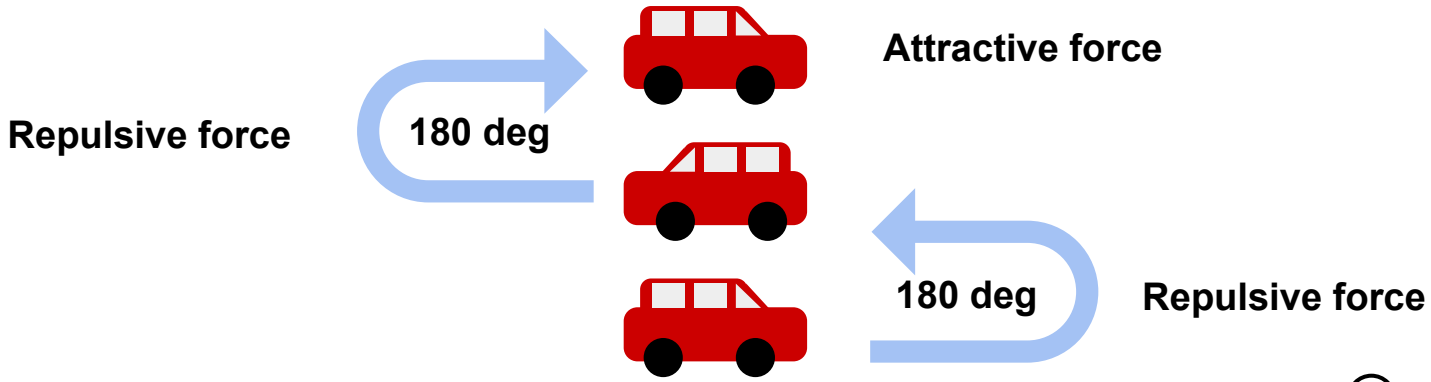


5. What is Electric Charge?

In R-space, electric charge is explained as a combination of rotation vector inversions.

An inversion is a 180-degree rotation of direction.

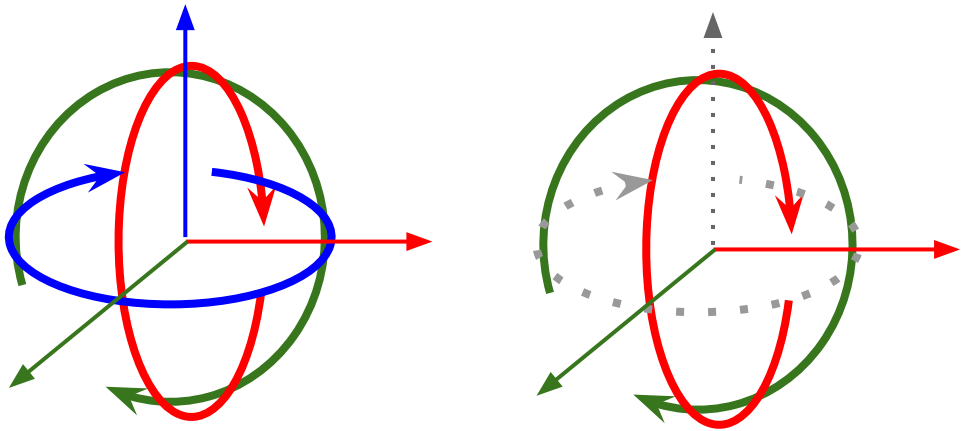
One 180-degree rotation causes repulsion, while two such rotations (returning to the original direction) result in attraction.



5. What is Electric Charge?

Electrons consist of three rotation vectors.

In contrast, as explained in later chapters, an up quark has only two rotation vectors because one component is confined.

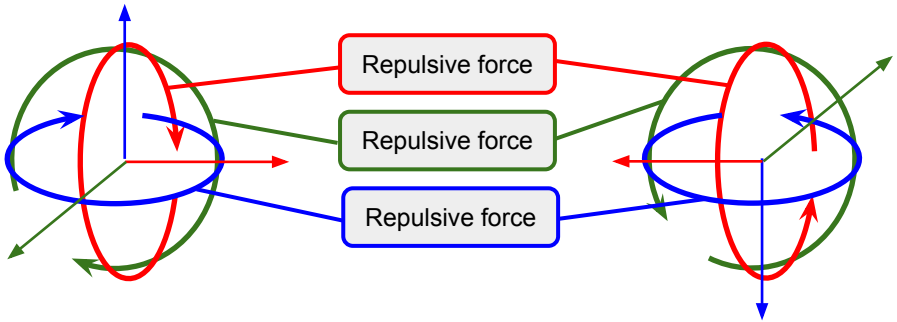


Electron **Up quark**

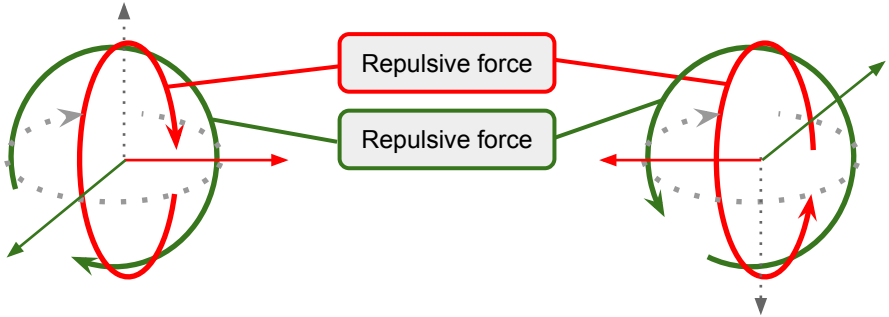
6. Interaction and the Charge Formula

When two electrons interact, their x, y, and z phases collide and move apart, resulting in repulsion.

Similarly, two up quarks experience repulsion as their x and y phases collide.



Two Electrons

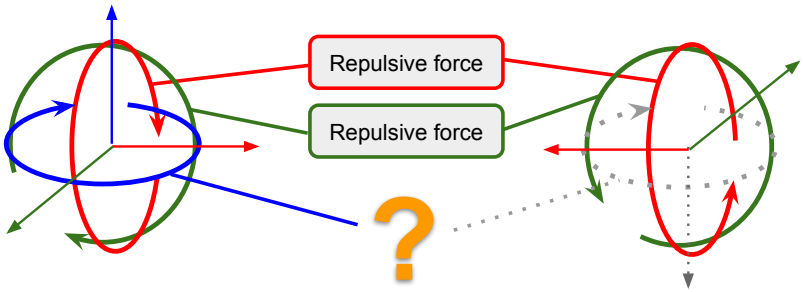


Two Up quarks

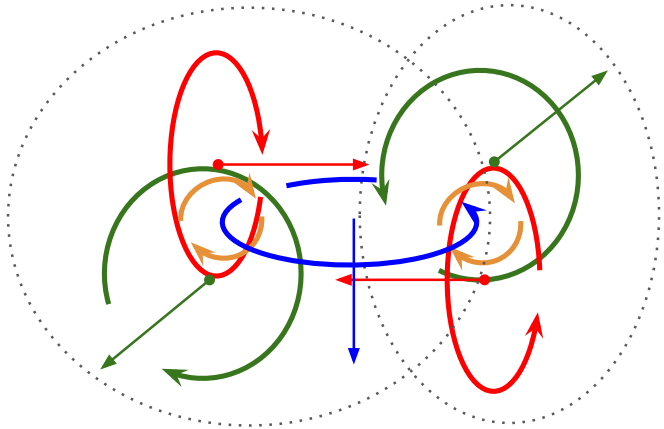
6. Interaction and the Charge Formula

However, in an electron-up quark interaction, the electron's x and y phases collide and repel, but its z-vector has no counterpart in the up quark.

Consequently, the electron's remaining z-vector pairs with the up quark's two-vector pair, causing them to orbit together.



Electron **Up quark**

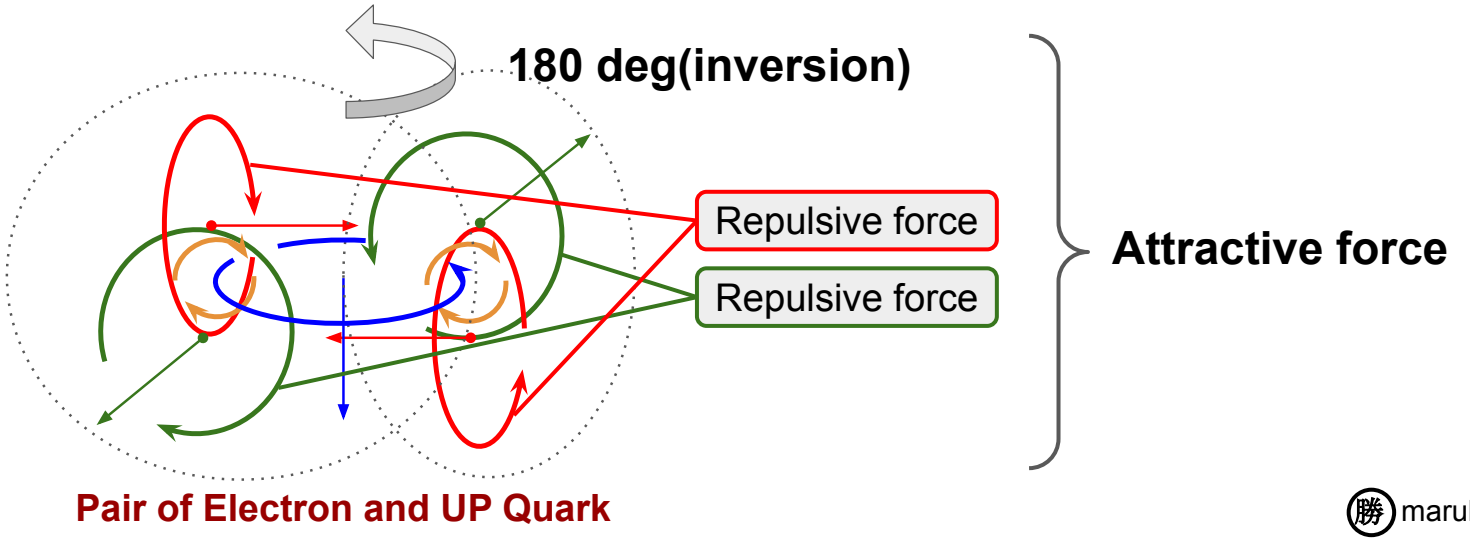


Pair of Electron and UP Quark

6. Interaction and the Charge Formula

Because the z-vector has only half the force required to rotate the entire system, it causes a 180-degree rotation (inversion).

This adds an extra inversion to the x and y components, totaling two inversions, which causes the phases to approach each other—this is the attractive force.



6. Interaction and the Charge Formula

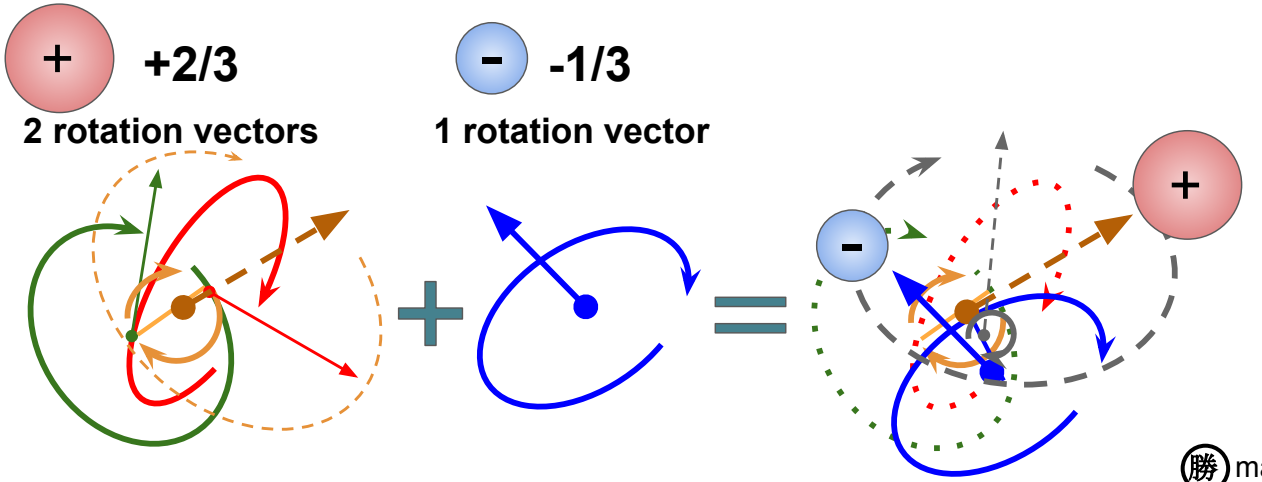
Given that rotation vectors represent angular momentum, the relationship between the number of rotation vectors (n) and the charge (q) is: q=(−1)^n · n/3.

$$q = (-1)^n \cdot \frac{n}{3}$$

7. Mechanism of Magnetism in Electron Spin

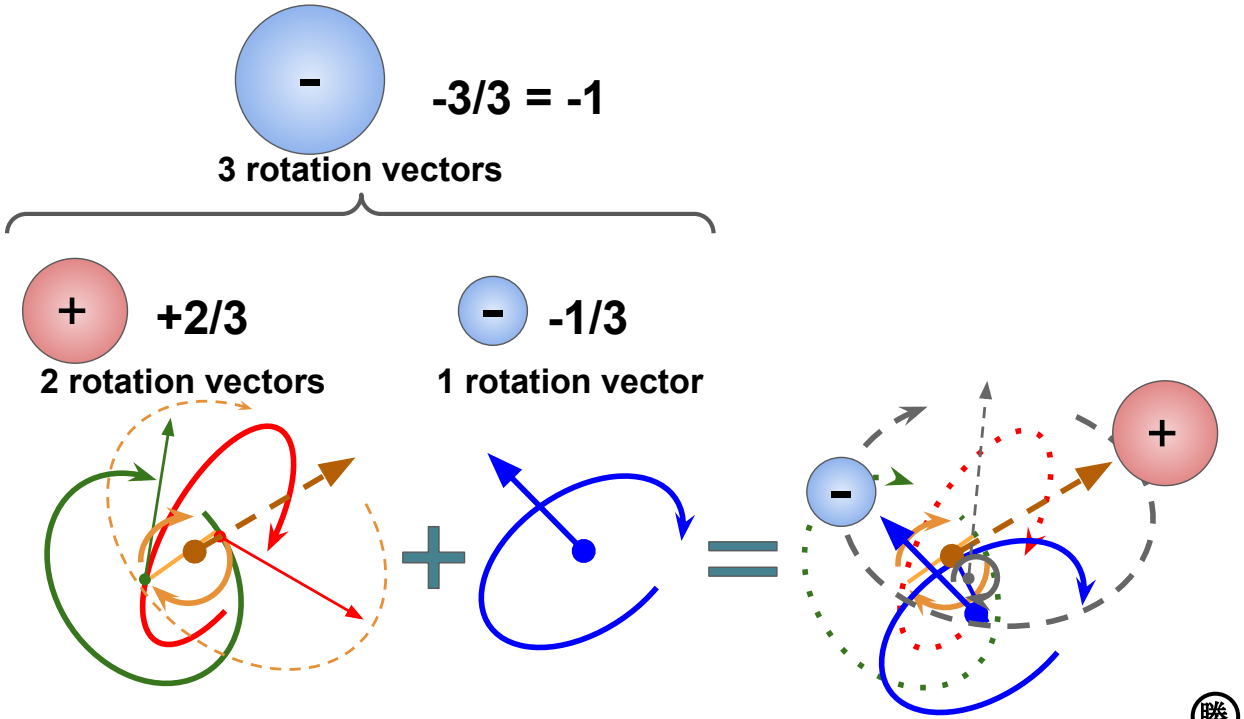
An electron consists of a pair of two rotation vectors and one single rotation vector orbiting each other.

Based on the charge formula, this is equivalent to a +2/3 charge and a -1/3 charge orbiting each other, which generates magnetic force like a microscopic electromagnet.



7. Mechanism of Magnetism in Electron Spin

However, the electron as a whole has a charge of -1 (calculated via inversions, not simple addition).

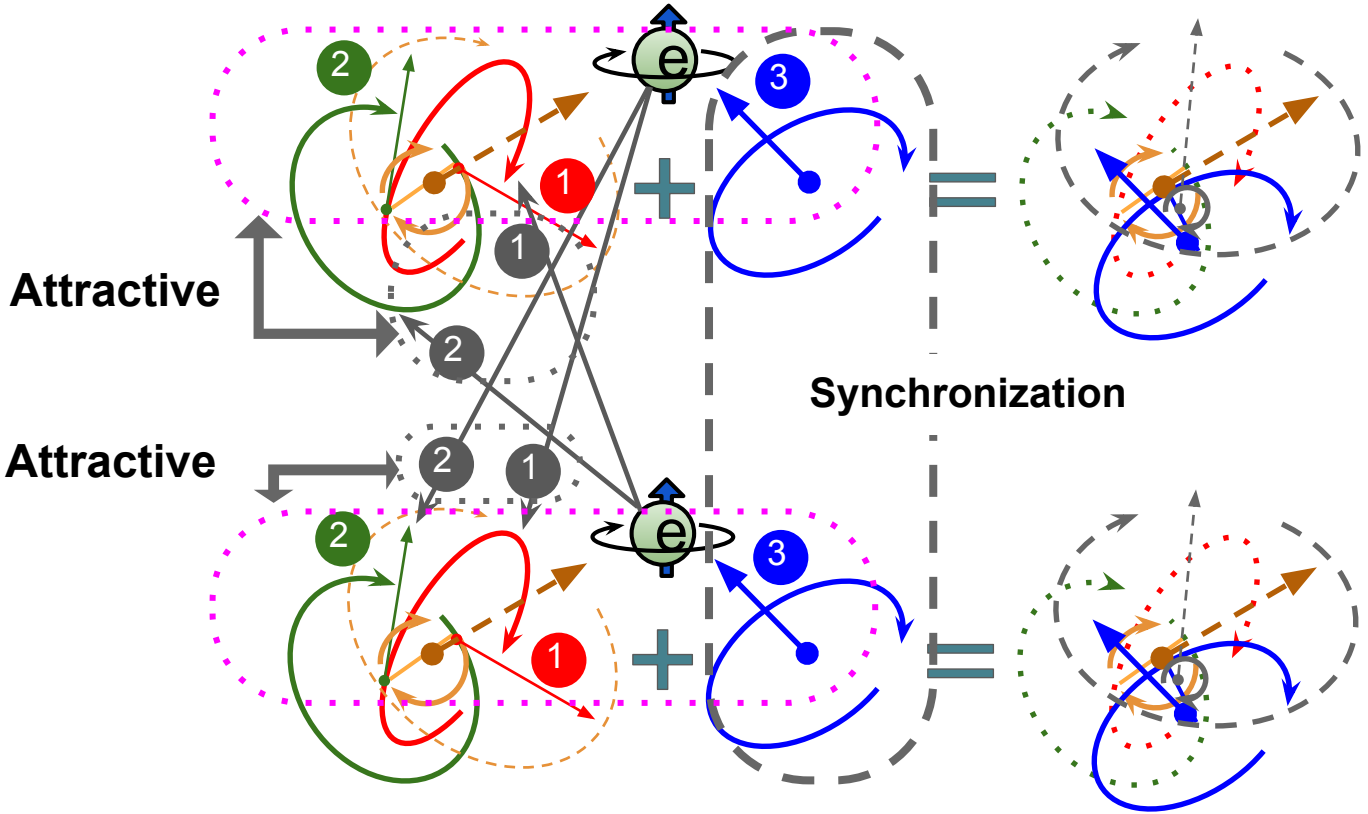


7. Mechanism of Magnetism in Electron Spin

Furthermore, magnetic force is not "emitted"; rather, electrons interact with each other.

When the spin axes and rotation directions of two electrons align, the precession of the partner appears to have only two components (x and y) due to synchronization, much like how a car moving at the same speed appears stationary.

7. Mechanism of Magnetism in Electron Spin



8. Conclusion

From each electron's perspective, it sees itself as having three vectors and the other as having two, causing their phases to approach each other.

In our world, this is perceived as magnetic attraction.

In summary, spin, charge, and magnetism all originate from rotation vectors and their precession.

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This is all for my presentation.

Thanks for watching!

